firstname lastname

email.com | xxx-xxx-1234 | linkedin.com | github.com | portfolio.com

EDUCATION

University of Pennsylvania, Philadelphia, PA | M.S.E. Robotics | GRASP Lab | Aug 2024 - May 2026
UPenn GRASP M.S.E. Robotics graduate focused on ROS 2 C++/Python perception systems for RGB-D mapping, terrain analysis, object tracking, SLAM/VIO evaluation, and manipulation. PointCloud2/GridMap/OccupancyGrid pipelines, RealSense/Gazebo validation, diagnostics, testing, and CI. | **Relevant coursework:** Deep Learning, Geometric and Advanced Perception, Computer Vision, Robot KInematics, Learning in Robotics

SKILLS

Robotics & Perception: ROS 2/ROS, C++, Python, TF/TF2, RViz, Gazebo/GZ Sim, ROS-GZ Bridge, PointCloud2, GridMap, OccupancyGrid, PCL, OpenCV
Sensors, Robots & SLAM: RealSense D455/L455, Franka Panda, RTAB-Map, ORB-SLAM3, OpenVINS, EuRoC MAV, KITTI, evo
ML & Software: PyTorch, NumPy, YOLOv8n, ByteTrack, TrackEval, pytest, gtest, GitHub Actions, Linux, Docker

EXPERIENCE

Robotics Lab Researcher, ModLab - University of Pennsylvania | Feb 2026 - May 2026
ROS 2, RealSense, Gazebo, PointCloud2, GridMap
• Built ROS 2 RGB-D terrain perception workflows for RealSense D455 and COMPA/HAMR Gazebo, converting live/sim streams into PointCloud2, elevation GridMap, PCD exports, and RViz diagnostics.
• Implemented rclpy/NumPy point-cloud cleaning, odometry transforms, 30 m x 30 m elevation maps, and slope/roughness traversability scoring; validated live/sim map growth with finite-cell checks.

PROJECTS

Real-Time Multi-Object Tracking for Robotics Perception | Python, C++, ROS 2, YOLOv8n, ByteTrack, KITTI
- Built YOLOv8n + ByteTrack pipeline with TrackEval HOTA/MOTA/IDF1 metrics, CPU/GPU latency benchmarking, dropped-frame robustness tests, failure diagnostics, and annotated demos.
- Added ROS 2 Jazzy replay/online smoke checks publishing detections, tracked objects, diagnostics, debug images, and risk/safety topics; implemented C++ tracking core with tests and microbenchmarks.

RGB-D Mapping and Terrain Perception Pipelines | ROS 2, C++/Python, RealSense, Gazebo, PCL, GridMap
- Developed mapping workflows across TUM replay, RealSense D455, and Gazebo L455; debugged CameraInfo/TF/PointCloud2 integration and generated C++/PCL maps, PCD/PLY exports, elevation GridMaps, and occupancy/traversability maps.
- Measured ~10 Hz replay and ~1.66 Hz map/grid updates; packaged parameter sweeps, launch/check scripts, tests, CI, and demos for repeatable validation.

Franka Panda Manipulation and Block Stacking | ROS, Python, Franka Panda, FK/IK, Null-Space Control
- Implemented Franka velocity/numerical IK using Jacobian pseudo-inverse/transpose methods, null-space joint centering, manipulability analysis, and RViz/hardware validation.
- Built block-stacking pipeline using object detections, camera-to-end-effector calibration, top-down grasp alignment, hover/drop strategies, and stack-height updates; achieved 100% successful attempted grasps on 4 static blocks in reported hardware tests.

VIO/SLAM Evaluation and Camera-IMU Calibration Diagnostics | Python, ROS 2, ORB-SLAM3, OpenVINS, EuRoC MAV, evo
- Automated ORB-SLAM3 vs. OpenVINS evaluation on EuRoC Machine Hall with TUM trajectory conversion, evo APE/RPE, fair-overlap analysis, failure-event extraction, plots, and error timelines.
- Ran OpenVINS calibration sensitivity study with frozen extrinsic/timestamp perturbations across 13 trajectories; quantified 25.82x ATE increase from a 5 deg camera-IMU z-axis rotation error.

ADDITIONAL PERCEPTION / ESTIMATION WORK:
AprilTag PnP/P3P pose recovery; two-view 3D reconstruction with essential matrix/RANSAC/triangulation; YOLO detector implementation; EKF parameter estimation; Bayes filtering; HMM smoothing.